Project Design Phase
Proposed Solution Template

Date	28 June 2026
Team ID	Not Given
Project Name	Rising Waters
Maximum Marks	2 Marks

Team Members	Skill Wallet Team ID's	Roll Numbers (College: ACET)
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Proposed Solution Template:

The proposed solution, Rising Waters, is a machine learning-based flood prediction system designed to provide accurate and timely flood risk assessments. The system analyzes historical weather data, including rainfall, cloud visibility, humidity, temperature, and seasonal rainfall, using algorithms such as Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost. The best-performing model is integrated into a Flask web application, enabling users and disaster management authorities to monitor flood risks through an intuitive interface. The solution supports early warning, efficient resource allocation, and informed decision-making while offering scalability through deployment on IBM Cloud.

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Floods often cause severe damage due to delayed warnings and inaccurate risk prediction. Existing systems lack localized, real-time analysis, making it difficult for authorities and citizens to prepare and respond effectively.
2.	Idea / Solution description	**Rising Waters** is an AI-powered flood prediction and early warning system that analyzes weather forecasts, rainfall, river water levels, and historical flood data to predict flood risks. It provides timely alerts and visual risk maps to help authorities and communities take preventive action.
3.	Novelty / Uniqueness	Integrates multiple real-time data sources with AI-based prediction to generate location-specific flood alerts. The system supports proactive disaster management through intelligent risk analysis and early warning notifications.
4.	Social Impact / Customer Satisfaction	Improves public safety by providing accurate early warnings, reducing loss of life and property. It helps disaster management authorities make informed decisions and increases community preparedness and confidence.
5.	Business Model (Revenue Model)	Subscription-based services for government agencies, municipalities, and disaster management organizations. Additional revenue through enterprise licensing, API integration, and customized analytics for organizations.
6.	Scalability of the Solution	The solution is cloud-based and can be expanded to multiple cities, districts, and countries. It supports integration with IoT sensors, weather APIs, satellite data, and mobile applications, making it suitable for large-scale deployment.